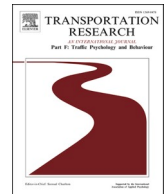




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Route learning with augmented reality navigation aids

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ABSTRACT

Current trends point towards the development and implementation of augmented reality (AR) navigation assistance systems for drivers. Previous research has found augmented reality can benefit driving performance, but the impact of AR on route learning (spatial knowledge acquisition) has received less attention. The current study used simulated driving scenarios to determine how the type of navigation aid device used (i.e., paper map, electronic map, and AR system) affected driving performance and route learning of 62 driving adults. Route learning was examined at landmark, route, and survey levels. The hypotheses tested were that AR would improve driving performance but diminish route learning compared to paper map use; and that AR would outperform electronic map use in both driving performance and route learning. Results indicated that paper map use led to poorer driving performance and greater route learning. However, map recognition may have been a confounding factor in higher level spatial knowledge acquisition with paper map use. Driving performance and route learning were similar for drivers using the electronic map and AR, but some differences suggested that drivers have reduced uncertainty and hesitation while using AR to navigate in unfamiliar areas.

1. Introduction

The development of the first automobiles in the late 19th century coincided with advancements in printing technology that allowed for a greater proliferation of maps among the general public (Library of Congress, 2019; Library of Congress, n.d.). For most of automotive history, paper maps have been the go-to option for drivers navigating an unfamiliar location. Although examples of personal navigation devices (PNDs) that update an egocentric representation of a driver's position have existed as far back as 1930, high costs and data storage limitations minimized their prevalence and usefulness on the road until relatively recently (Leite, 2018). Conventional PNDs, such as GPS devices and smartphone apps, have become increasingly common over time as technological advancements reduced cost and increased effectiveness. A 2015 survey by Kamalanathsharma et al. (2015) found that 64% of drivers use an app for navigation or traffic information, and a 2011 study by Lo et al. found frequent PND use even when the destination was familiar to the driver. High PND usage is not surprising considering its effectiveness in increasing users' driving performance and wayfinding capabilities in unfamiliar areas (Lee & Cheng, 2008; Streeter et al., 1985), leading to more efficient driving that ultimately results in net savings of billions of dollars and massively reduced CO₂ emissions (O'Connor et al., 2019).

One avenue of consideration in the continuing development of PNDs is augmented reality (AR) technology, which presents a direct or indirect view of the environment with overlaid computer-generated information or images (Azuma, 1997; Carmigniani et al., 2011; Lee, 2012). Studies investigating AR HUDs (heads-up displays) that use the vehicle's windshield to display information directly in the

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driver's field of view have found benefits compared to the use of conventional heads-down display (HDD) PNDs, such as enhanced response times (Bolton et al., 2015) and less time spent looking away from the road (Bark et al., 2014; Medenica et al., 2011). AR HUDs also have the potential to reduce stress in drivers with high levels of interpersonal anxiety through hazard identification (Hwang et al., 2016). Additionally, AR HUDs do not necessarily appear to impose a distraction. For example, Rusch et al. (2013, 2014) did not observe interference resulting from the presence of AR cues, and Smith et al. (2017) found that HUDs afforded longer glances without negatively affecting vehicle control and supported higher levels of task complexity compared to conventional HDD PNDs. These benefits taken together with a driver preference for HUD and AR systems over HDD systems (Jose et al., 2016; Kim & Dey, 2009; Medenica et al., 2011), and continuing technological advancements that increase the likelihood of a windshield-based AR HUD PND being developed (Gabbard et al., 2014), suggest that AR HUD PNDs will become more prevalent on the road in the future. With these considerations, an examination of the impact of AR PND use compared to conventional HDD style PND use is warranted.

Route learning in particular is one factor for which the impact of AR HUD PND use is not fully understood. Also known as spatial knowledge acquisition, route learning refers to a person's ability to form a cognitive map of an area. The construction of cognitive maps is important because it persists beyond the immediate period of PND use, reducing device dependency and allowing unassisted navigation during times when a driver's PND is unavailable or incapable of providing effective instruction. The formation of cognitive maps can be divided into three levels with increasing degrees of comprehensiveness, complexity, and usefulness to navigation. The levels are landmark knowledge, which includes individual locations and objects without relation between them; route knowledge, in which landmarks are linked in order; and survey knowledge, in which multiple routes are interlinked as a greater spatial understanding (Siegel & White, 1975; Stern & Leiser, 1988). Although the highest levels of route learning are not required to reach a destination, survey knowledge of an area allows a navigator to more effectively correct errors, react to situational demands, and develop alternative paths to their destination.

In contrast to the largely beneficial impact of PND use on driving performance and wayfinding, the impact of traditional PND use on route learning appears much less positive. In general, greater cognitive effort leads to greater spatial recall (Tyler et al., 1979). Conventional PND usage reduces the cognitive requirements of navigation for drivers, potentially leading to inattentional blindness and diminishing their ability to retrace a route without assistance (Fenech et al., 2010). Compared to drivers using a paper map, drivers using conventional PNDs tend to acquire lower levels of spatial knowledge, including distortions and omissions in their mental representation of the environment (Dickmann, 2012; Ishikawa & Takahashi, 2013; Münzer et al., 2012; Rizzardo et al., 2013; Waters & Winter, 2011). There is also evidence to suggest that high PND use and dependency negatively impacts spatial memory as a whole and is not limited to only affecting the level of route learning for the areas in which PND use occurs (Dahmani & Bohbot, 2020). An active focus on navigation is more significant than the act of driving itself with regards to route learning, as seen in the greater degree of route learning observed in "backseat drivers" who take charge of navigation compared to the passive drivers they direct (von Stülpnagel & Steffens, 2012). This finding suggests that any method of navigation or navigation assistance must have a degree of involved effort on the part of the navigator in order to develop a robust level of spatial knowledge, and that the differences between navigation assistance systems can unintentionally impact users' route learning.

While paper maps, conventional PNDs, and AR systems can all provide the same fundamental information about the route to be traveled through the environment, they differ in the presentation of this information, which has implications for the degree of cognitive effort requirements and thus their facilitation of route learning. Compared to using a paper map, using a PND requires less active focus and simplifies navigation. Whereas paper map use requires users to orient themselves within the spatial representation of their environment, account for heading, locate turns, etc., PNDs perform these functions automatically. The more direct incorporation of navigation instructions into the environment and enhanced attention associated with AR HUD systems suggests the potential for enhanced route learning compared to conventional PNDs.

This study utilized a simulated driving task to examine the impact of using a hypothetical AR HUD navigation assistance system compared to an electronic map (emap) and a paper map. The AR HUD system displayed directional road overlays and arrows in the driver's field of view to indicate turn locations. The emap was designed to emulate the style of conventional navigation assistance systems (e.g., HDD heading-oriented map with audio indication of turn locations) in an effort to explore potential changes to route learning in drivers resulting from developments in PND technology. The paper map served as a control condition in which participants had access to information concerning the driving route and environment without the turning instructions provided in the AR and emap conditions. Measures of route learning and driving performance were collected. We hypothesized that the best driving performance would coincide with AR use. We also hypothesized that the greatest degree of route learning would occur with paper map use due to the greater levels of cognitive effort required by paper map usage, and that route learning when using AR would be greater than when using the emap.

2. Methods

2.1. Participants

Undergraduate psychology students participated in this study through a research participant pool for course credit. Out of an initial 64 participants, two did not complete all three driving conditions, leaving 62 participants (13 men, 49 women). Data were excluded from route learning analyses if the participant did not complete the online follow-up survey, or if they committed one or more driving errors that resulted in at least one landmark being skipped, failing to load, or otherwise not appearing during any of the driving sessions. Additionally, the maps used to assess survey knowledge were changed after the first nine participants. Data from those participants were excluded for map selection errors analysis. Due to overlap between these exclusion criteria, data from 23 participants

were used in analyses of sensory efficiency and landmark ordering errors, and data from 18 participants were used in the analysis of map selection errors. Data from all 62 participants were used for analyses of driving performance.

Participants ranged in age from 18 to 59 years ($M = 23$) with a median age of 21 years. All participants possessed a valid driver's license and normal or corrected to normal vision (20/30 or better, as assessed by a Snellen eye chart). The average number of years of driving experience for participants was 6.78 ($SD = 7.10$) years. Of the 62 participants whose data were analyzed, 19.4% self-identified as Black or African American ($n = 12$), 67.7% as White ($n = 42$), 9.7% as Hispanic or Latino ($n = 6$), 1.6% as Asian or Pacific Islander ($n = 1$), and 1.6% as Other ($n = 1$). Additional information about participant driving history and navigation aid device use is summarized in Table 1.

2.2. Design and procedure

The study employed a within-subjects design to investigate the impact of the use of navigation aid devices while driving on route learning and driving performance. Latin square design was used when assigning condition order.

2.2.1. Laboratory appointment

Participants' vision was assessed using a Snellen eye chart, then participants completed a survey consisting of basic demographics information and driving history (see Table 1).

In order to reduce potential practice and order effects, participants completed a practice drive lasting approximately 5 min. The practice drive was followed by three separate simulated driving sessions lasting approximately 5 min each. Each driving session utilized one of three navigation aid conditions.

All driving was completed using a STISIM® driving simulator. The simulator included 3 large LCD panel displays providing an approximately 155 degree field of view, including a rear view and two side view simulated mirrors. The steering wheel and accelerator and brake pedals were highly realistic in look and feel. Participants were instructed to drive as if they were driving normally in a real car. The driving scenario for each of the three driving sessions was constructed from the same base course, but the order of appearance of course sections was rearranged and some features were reoriented between conditions in order to reduce practice effects.

The different conditions during the driving sessions were as follows:

- Paper Map: participants were allowed to study the map of the course (see Fig. 1) for as long as they wished. In addition to studying the map prior to driving, participants were able to reference the map while driving during the paper map condition.
- Emap: A small, nose-up map was located on the dash and continually updated based on the driver's position (see Fig. 2). Vocal instructions were issued by the simulator to indicate turns (e.g., at the next intersection, turn right).
- AR: Arrows were displayed on the screen 12ft above ground level at turn locations. A dashed green line was placed on the road leading up to turn locations (see Fig. 3).

Table 1
Driving history and navigation aid device use.

	<i>n</i>	%
Navigation aid device use when driving anywhere		
Never	1	1.6
Sometimes	36	58.1
About half the time	11	17.7
Most of the time	14	22.6
Always	0	0
Navigation aid device use when driving to an unfamiliar situation		
Never	0	0
Sometimes	1	1.6
About half the time	2	58.1
Most of the time	21	17.7
Always	38	22.6
Type of navigation aid device used		
Feature of the car	3	4.8
Separate, dedicated navigation device	4	6.5
Mobile app	55	88.7
Visual information presentation method for navigation aid device		
Map (north-up)	2	3.2
Map (nose-up)	57	91.9
Street view	3	4.8
Audio information presented by navigation aid device		
Yes	54	87.1
No	8	12.9
At-fault accident involvement		
Yes	22	35.5
No	40	64.5

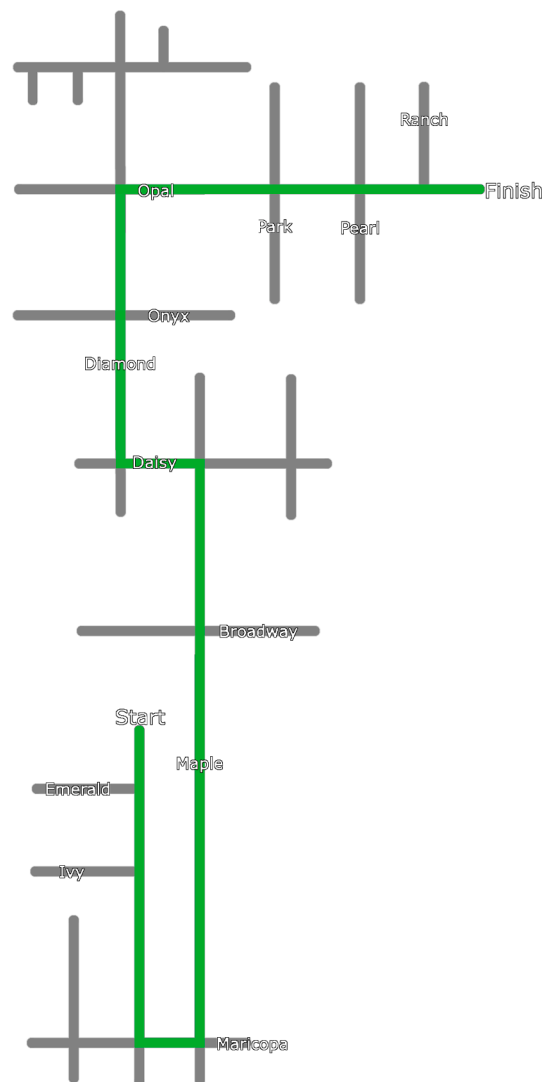


Fig. 1. Paper map condition. Participants received a printed, laminated copy of the map scaled to the maximum possible size on 8.5×11 " paper.



Fig. 2. Emap condition. The green line indicates the intended route and the map updates to maintain the driver's heading as the upward position. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)



Fig. 3. AR condition. The dashed green line and arrows indicate the intended route. A left turn is indicated at the next intersection. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

After each driving session, participants completed a series of questions adapted from [Burnett and Lee \(2005\)](#) in order to assess their landmark, route, and survey knowledge:

- Landmark knowledge was assessed by presenting a series of images taken from the intended driving route alongside images which were not part of the driving route. Participants were asked to identify which of the images were part of the route they drove. Pictures were presented in a set of ten — eight landmarks that were present in the drive, and two distractor landmarks that were not present. Participants' hits and false alarms were recorded.
- Route knowledge was assessed by asking participants to arrange the landmarks selected during landmark knowledge assessment according to their order of appearance during the drive. Errors in placement and landmark selection were recorded.
- Survey knowledge was assessed by asking participants to identify the map of the route they drove from a selection of ten maps. One of the maps perfectly matched the course while the other nine maps each contained a unique number of errors ranging from 1 to 9. The number of errors present in the selected map were recorded. This differs from [Burnette and Lee \(2005\)](#) in which participants were asked to draw a map from memory with as many features as possible, which researchers later coded for accuracy to the course.

Following the completion of all three driving sessions and post-drive tests, participants were asked if they wished to continue participation in the online follow-up survey one week later. Participants who declined to participate in the follow-up survey were debriefed.

2.2.2. Online follow-up survey

In order to assess spatial memory retention, participants completed an online survey one week after completing the lab visit. Questions on the follow-up survey were identical to the post-drive questions completed during the lab visit (see 2.2.1).

2.2.3. Analysis

Data analysis was performed using SPSS. Measures of driving performance (signal light usage, lane excursions, crashes, speeding, completion time, turning errors, and missed stops) were collected by the driving simulator and the effect of navigation aid device type on those measures was analyzed using one-way repeated measures ANOVAs. A Greenhouse-Geisser correction was applied for variables that violated the assumption of sphericity: centerline crossings and road excursions, total crashes, and number of turning errors.

Measures of route learning were collected at the landmark level (sensory efficiency for landmark recognition), route level (proportion of error per landmark selection), and survey level (number of errors in the selected map) during the post-drive questionnaires during the lab visit (see 2.2.1) and the one week follow-up questionnaire (see 2.2.2). Errors in the selected map did not require further calculation. Sensory efficiency and error proportion were calculated as follows:

- Sensory efficiency was calculated using the formula developed by Green and Swets (1966): $A_G = [P(H) + \{1 - P(FA)\}] / 2$, where A_G = Sensory Efficiency, an estimated measure of the area under the receiver operating characteristics (ROC) curve; $P(H)$ = hit rate; and $P(FA)$ = false alarm rate. Sensory efficiency (A_G) is a scalar quantity ranging from 0 to 1.0 and is strongly correlated to sensitivity index (d'), a measure of an individual's ability to detect a signal (Craig, 1979).
- Proportion of errors per landmark was calculated by dividing the number of errors made (number of landmarks placed early, number of landmarks placed late, and number of false positive landmarks) by the number of landmarks selected.

Two-way repeated measures ANOVAs were used to analyze the effects of navigation aid type (paper map, emap, AR) and test time (post-drive immediate, post-drive one week later) on route learning measures.

3. Results

3.1. Driving performance

See Table 2 for descriptive statistics and one-way ANOVA results for driving performance measures.

Post hoc testing using the Bonferroni correction was conducted to assess differences between navigation aid conditions.

The number of good usages of signal lights during turns and lane changes did not significantly differ between emap usage and AR usage ($p = 0.50$). However, the number while using a paper map was significantly less than the number while using the emap ($p < 0.001$) and AR ($p < 0.001$).

The number of poor and missed usages of signal lights during turns and lane changes did not significantly differ between paper map usage and AR usage ($p = 0.078$), and the number did not significantly differ between emap usage and AR usage ($p = 1.00$). However, the number was significantly greater while using the paper map than while using the emap ($p < 0.05$).

Total drive time did not significantly differ between paper map usage and emap usage ($p = 1.00$). However, the total drive time while using AR was significantly less than the time while using the emap ($p < 0.05$) and paper map ($p < 0.05$).

The number of poor signal usages during turns did not significantly differ between paper map usage and emap usage ($p = 1.00$). However, the number while using AR was significantly less than the number while using the paper map ($p < 0.05$) and emap ($p < 0.05$).

The number of poor signal light usages during lane changes did not significantly differ between paper map usage and AR usage ($p = 0.96$). However, the number of poor turn signal usages during lane changes while using the emap was significantly less than the number when using both the paper map ($p < 0.05$) and AR ($p < 0.01$).

The number of turning errors did not significantly differ between emap usage and AR usage ($p = 0.50$). However, the number of turning errors when using the paper map was significantly greater than the number when using both the emap ($p < 0.001$) and AR ($p < 0.001$).

No statistically significant differences were found between navigation aid types for any other driving performance variable.

3.2. Route learning

See Table 3 for descriptive statistics and two-way ANOVAs of the effects of navigation aid device type and time of testing on route learning.

There were no statistically significant interactions for sensory efficiency ($p = 0.252$), landmark ordering errors ($p = 0.968$), or map selection errors ($p = 0.539$). There were statistically significant main effects of time on sensory efficiency, $F(1, 22) = 16.12$, $p < 0.01$, $\eta_p^2 = 0.42$, landmark order errors, $F(1, 22) = 7.97$, $p < 0.05$, $\eta_p^2 = 0.27$, and map selection errors, $F(1, 17) = 19.28$, $p < 0.001$, $\eta_p^2 = 0.53$. Sensory efficiency decreased between the lab visit (0.74 ± 0.02) and the follow-up (0.64 ± 0.03), landmark ordering errors increased between the lab visit (0.54 ± 0.05) and the follow-up (0.70 ± 0.05), and map selection errors increased between the lab visit (2.26 ± 0.38) and the follow-up (4.52 ± 0.39).

Table 2

Descriptive statistics and one-way ANOVA for driving performance.

Measure	Paper <i>M (SD)</i>	Emap <i>M (SD)</i>	AR <i>M (SD)</i>	<i>F</i>	<i>df</i>	η_p^2
Good Signal Use	3.65 ^a (1.64)	4.45 ^b (1.71)	4.69 ^b (1.20)	18.29***	2, 122	0.23
Poor & Missed Signal Use	4.63 ^a (3.00)	3.90 ^b (3.20)	3.84 ^{ab} (2.32)	3.68*	2, 122	0.06
Centerline Crossings & Road Excursions	5.55 (2.24)	5.56 (1.25)	5.79 (1.44)	0.49	1, 55, 94.36	0.01
Total Crashes	0.05 (0.22)	0.05 (0.28)	0.00 (0.00)	1.29	1, 72, 105.01	0.02
Total Drive Time	324.41 ^a (50.54)	322.61 ^a (47.42)	311.40 ^b (41.90)	5.17**	2, 122	0.08
Poor Signal Use (Turns)	0.66 ^a (0.90)	0.58 ^a (0.82)	0.31 ^b (0.53)	5.41**	2, 122	0.08
Poor Signal Use (Lane Changes)	1.61 ^a (1.03)	1.13 ^b (1.08)	1.79 ^a (1.04)	7.14**	2, 122	0.10
Number of Speeding Incidents	3.48 (1.82)	3.45 (2.07)	3.76 (2.04)	1.00	2, 122	0.02
Percent of Time Spent Speeding	18.46 (10.53)	19.21 (11.69)	19.67 (11.83)	0.61	2, 122	0.01
Percent of Distance Spent Speeding	28.04 (15.31)	28.54 (16.68)	28.29 (16.46)	0.05	2, 122	0.00
Number of Turning Errors	0.90 ^a (1.22)	0.08 ^b (0.45)	0.00 ^b (0.00)	28.20***	1, 2, 73.42	0.32
Number of Missed Stops	2.42 (1.43)	2.55 (1.54)	2.27 (1.67)	1.13	2, 122	0.02

Note. $n = 62$; Time measured in seconds; ANOVA = analysis of variance; Superscript a, b represents post-hoc comparison among navigation aid device types. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 3
Descriptive statistics and two-way ANOVA for route learning.

Variable		Paper Map		Emap		AR		ANOVA			
		<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	Effect	<i>F</i>	<i>df</i>	η_p^2
A_G	T_1	0.69 ^{ay}	0.20	0.75 ^{ay}	0.12	0.78 ^{ay}	0.10	D	3.17	2, 44	0.13
	T_2	0.61 ^{aby}	0.23	0.59 ^{az}	0.18	0.71 ^{bz}	0.16	T	16.12**	1, 22	0.42
LOE	T_1	0.51 ^{ay}	0.39	0.50 ^{ay}	0.41	0.62 ^{ay}	0.43	D × T	1.42	2, 44	0.06
	T_2	0.65 ^{ay}	0.45	0.67 ^{ay}	0.40	0.77 ^{ay}	0.41	D	1.07	2, 44	0.05
M_{err}	T_1	1.22 ^{ay}	0.65	2.72 ^{by}	2.49	2.83 ^{by}	2.71	T	7.97*	1, 22	0.27
	T_2	3.67 ^{az}	2.77	4.33 ^{abz}	2.77	5.56 ^{bz}	2.18	D × T	0.03	2, 44	0.07
								D	6.70**	2, 34	0.28
								T	19.28***	1, 17	0.53
								D × T	0.63	2, 34	0.04

Note. A_G = Sensory Efficiency; LOE = Landmark Ordering Errors; M_{err} = Map Selection Errors; $n = 23$ for A_G and LOE; $n = 18$ for M_{err} ; D = navigation aid device type; T = timing of test; ANOVA = analysis of variance; T_1 = post-drive test; T_2 = follow-up test; Superscript a, b represents simple effects comparison among navigation aid device types; Superscript y, z represents simple effects comparison among times. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

There was a significant main effect of navigation aid device type on map selection error, $F(2, 34) = 6.70$, $p < 0.01$, $\eta_p^2 = 0.28$. Post-hoc testing using the Bonferroni correction revealed that paper map use (2.44 ± 0.32) and AR use (4.19 ± 0.34) significantly differed in the number of errors present in the selected map ($p < 0.01$). There was not a significant difference between emap use (3.53 ± 0.51) and paper map use ($p = 0.21$) or between emap use and AR use ($p = 0.48$).

4. Discussion

4.1. Driving performance

The hypothesis that AR would lead to the best driving performance and the paper map would lead to the worst was partially supported by the results — drivers using AR outperformed drivers using the paper map, but there was not a clear difference in driving performance between the emap and AR conditions. The number of turning errors made was the clearest factor with almost all erroneous turning actions being completed while using the paper map, a small amount while using the emap, and none while using the AR system. This result, like many of the differences observed in driving performance, can likely be attributed to unfamiliarity with using paper maps and north-up oriented navigation aids. In the words of one participant: “I’m used to it telling me where I’m supposed to go.” The extra demands of using the paper map, such as the necessity of performing mental rotations in order to determine turning direction, were removed by both the emap and AR system.

In terms of signal light usage, the paper map led to the poorest performance overall with a significantly lower number of good signal usages and a significantly higher number of poor usages compared to both the emap and AR conditions. Although drivers demonstrated better signal use when using both the emap and AR compared to their use of the paper map, the areas of improvement were different between them. Participants signaled turns better while using AR, but they signaled lane changes better while using the emap. This difference in signaling behavior would suggest that drivers using AR are better able to anticipate the locations of turns, but place less focus on the maneuvers necessary to reach those turns.

Participants completed the course faster while using AR than when using the paper map and emap, but the difference in driving time was small — only about 10 s, a reduction in total drive time of <4%. Interestingly, the amount of speeding did not significantly differ among conditions when viewed in terms of the total number of speeding incidents, percent of time spent speeding, and percent of distance spent speeding. This finding indicates that, rather than increased speeding causing the shorter time when using AR, participants had periods of slowing down that led to a longer time while using the paper map and emap.

Although the initial comparisons of AR and the emap seem to indicate little difference between AR and conventional navigation aid devices, AR does present some advantages in terms of driving performance. The greater constancy in speed combined with the reduction in poor signaling for turns while using AR suggests that instructions presented via AR are easier to interpret and follow compared to information displayed by conventional navigation aid devices similar to the emap. Differences in the ease of interpreting and implementing information have important implications for real world driving scenarios where uncertainty regarding instructions can lead to negative outcomes such as missed turns, disrupted traffic, or crashes. Even in situations where uncertainty does not lead to overt external consequences, the potential for AR to reduce stress for drivers navigating an unfamiliar area for the first time is a notable consideration.

4.2. Route learning

Hypotheses related to route learning — that drivers in the AR condition would outperform drivers in the emap condition, and that drivers in the paper map condition would outperform drivers in both the AR and the emap conditions — were partially supported. Sensory efficiency and landmark ordering errors, which measured landmark knowledge and route knowledge, respectively, were not observed to differ among navigation aid conditions. These results indicate that AR and the emap facilitated comparable degrees of

spatial knowledge acquisition at the landmark and route level.

Participants had fewer errors in map selection, which measured survey knowledge, when using the paper map compared to using AR. While the lower number of errors seen with paper map use appears to fit with past findings that frame navigation aids as inferior to studying a paper map when it comes to learning and remembering a route (Huang et al., 2012; Münzer et al., 2006; Rehman & Cao, 2017), caution is warranted in attempting to describe the lower number of errors in map selection for the paper map condition as an indicator of superior survey level knowledge. The difference between the paper map and AR was the only difference observed among navigation aid devices — the paper map and emap were not significantly different, nor were the emap and AR. One possible interpretation of these results is that AR and the emap support acquisition of lower levels of spatial knowledge to the same degree as studying the paper map, but are inferior for acquiring survey level knowledge. It should be noted, however, that it is unlikely that survey level knowledge was acquired during a single drive lasting roughly five minutes.

In contrast to driving performance, AR does not appear to offer any definitive benefits for route learning. Individuals who are concerned with the speed or depth of spatial knowledge acquisition, such as individuals interested in employment in emergency services, should not use navigation aid devices, AR or otherwise, while attempting to familiarize themselves with an area. On the other hand, AR systems would be highly beneficial to any driver who needs to travel to an unfamiliar location with little time to prepare or learn the area, especially in busy areas where distraction or uncertainty can have a larger impact while driving. User experience should also be taken into consideration. Driver preference for AR systems over conventional systems (Kim & Dey, 2009; Medenica et al., 2011) becomes a major factor when there are few, if any, downsides to consider when deciding between conventional systems and hypothetical AR systems.

4.3. Potential limitations and future directions

The difficulty of the driving courses was low, which may have made it difficult to assess some effects. For example, there were no significant differences observed in the number of crashes, but the average number of crashes was zero or nearly zero in all conditions. There were very few opportunities for participants to crash without engaging in extremely reckless driving behaviors. Similarly, the durations of the driving sessions were short in comparison to what a person might expect to experience in a real-world setting. This study attempted to minimize participant fatigue by keeping the driving sessions short, but future research could benefit from utilizing more difficult, real-world conditions that might exacerbate limitations in driving performance.

Due to the nature of the simulator, participants had to be removed from route learning analyses if they made an error that resulted in missing a landmark. The simulator loads features of the course in a linear manner based on distance and does not allow backtracking, which means that mistakes can cause alterations to the geometry of the course, and that participants will eventually be placed back on the “correct” path regardless of their turning behavior or attempts to self-correct their course. Future research should use a simulator that loads the entirety of the driving environment, allowing participants to explore and navigate without limitations.

A convenience sample of undergraduate psychology students was used for this study. For the most part, they were young, healthy individuals with average or better spatial memory and <10 years of driving experience. A more representative sample should be used in the future to increase generalizability of the current findings. This could be done by including older adults and individuals with varying levels of driving experience and spatial memory. Diversifying the sample would help to elucidate any relationships involving spatial memory that may not have been apparent as a result of the methodology used in this study. Additionally, a wider range of ages among participants would also allow for the examination of driving when AR is used by older drivers. The effects of AR on route learning may differ between older and younger drivers in a manner similar to its differential effects on driving performance compared to conventional systems. Although driving performance is enhanced for both older and younger drivers when using an AR system compared to using a conventional in-car system on a separate screen, the difference is far more pronounced for older drivers (Kim & Dey, 2009; Mourant et al., 2001).

The construct validity of map selection errors as a measure of survey knowledge is questionable. The study from which route learning questions were adapted, Burnett and Lee (2005), asked participants to draw a map with as many features as possible in order to assess survey knowledge. This study instead asked participants to pick a map from a selection of ten. Although this alteration reduced the time demands on participants, the potential for errors and non-responses in the follow-up, and the difficulty of coding, this method tested recognition, rather than recall, in the formation of a mental map during testing of the paper map condition.

CRediT authorship contribution statement

Zachary F. Yount: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Steven J. Kass:** Conceptualization, Methodology, Formal analysis, Resources, Writing – review & editing. **James E. Arruda:** Conceptualization, Methodology, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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